

Title	Oak tree leaf budding phenology, Silwood Park UK
General metadata	
Abstract	The leaf budding phenology of a proportion of tagged oak trees (<i>Quercus robur</i>) is annually monitored from March to June to register the date of five stages from bud burst to fully extended and hardened leaves.
Keywords	budburst, oak, phenology, food chain, climate change
Links	https://www.imperial.ac.uk/silwood-park/ https://doi.org/10.5281/zenodo.14771428
DOI	10.5281/zenodo.14771428
Is this part of a larger study?	Yes, these observations are part of the long-term monitoring of oaks and blue tit matching phenology study
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Funding	Imperial College London, Department of Life Science
Data set status and accessibility	
Status	Ongoing
Latest update	February 2025
Latest archive date	February 2025
Metadata status	February 2025
Accessibility	
Storage location and medium	"Research group space: SilwoodLTE", Imperial College London, ICT department
Usage rights	Restricted
Usage rights	Most is restricted but phenology data from 2007-2019 is available in https://doi.org/10.6084/m9.figshare.11856108.v4
Metadata	Zenodo (https://doi.org/10.5281/zenodo.14771428)

Geographic metadata																											
Geographic description	The study site is Silwood Park Campus from Imperial College London, Buckhurst Road, Ascot, Berkshire SL5 7PY, United Kingdom. Silwood Park campus, with about 70 ha, contains ancient woodlands and few-decades-old oak-dominated woodlands. Study oak trees have been set across the campus woodlands, which are classified as W10a, W10e and W16a using the National Vegetation Classification. Silwood Park experiences an average annual rainfall of 698mm with little seasonal pattern (1987-2022). Mean hourly temperature is 10 °C with July max of 23 °C and January min of 1.4 °C (1987-2022) (https://zenodo.org/records/14680305)																										
Bounding coordinates	General for Silwood Park																										
Latitude	51.411																										
Longitude	-0.647																										
UK National grid																											
Square	SU																										
Easting	94196																										
Northing	68866																										
Temporal metadata																											
Temporal description	Oak leaf bursting has been observed in a series of trees since 2007. Trees have been monitored each year (Long term trees) or every other year (even and odd trees trees) Trees enter the database or change the frequency of monitoring to replace dead individuals. Missing data: 2020																										
Begin	2007																										
End	Ongoing																										
Taxonomic metadata																											
General Information																											
Taxonomic level:	Angiospermae																										
Taxonomic level: Species	<table> <tr> <td colspan="2">Table: NAMESP</td></tr> <tr> <td>Species</td><td>Species code</td></tr> <tr> <td><i>Quercus robur</i></td><td>quercus.robur</td></tr> <tr> <td><i>Fagus sylvatica</i></td><td>fagus.sylvatica</td></tr> <tr> <td><i>Populus sp.</i></td><td>populus.sp</td></tr> <tr> <td>Unidentified conifer</td><td>conifer</td></tr> <tr> <td><i>Castanea sativa</i></td><td>castanea.sativa</td></tr> <tr> <td><i>Carpinus betulus</i></td><td>carpinus.betulus</td></tr> <tr> <td><i>Betula pendula</i></td><td>betula.pendula</td></tr> <tr> <td><i>Alnus sp.</i></td><td>alnus.sp</td></tr> <tr> <td><i>Aesculus hippocastanum</i></td><td>aesculus.hippocastanum</td></tr> <tr> <td><i>Acer pseudoplatanus</i></td><td>acer.pseudoplatanus</td></tr> <tr> <td><i>Juglans regia</i></td><td>juglans.regia</td></tr> </table>	Table: NAMESP		Species	Species code	<i>Quercus robur</i>	quercus.robur	<i>Fagus sylvatica</i>	fagus.sylvatica	<i>Populus sp.</i>	populus.sp	Unidentified conifer	conifer	<i>Castanea sativa</i>	castanea.sativa	<i>Carpinus betulus</i>	carpinus.betulus	<i>Betula pendula</i>	betula.pendula	<i>Alnus sp.</i>	alnus.sp	<i>Aesculus hippocastanum</i>	aesculus.hippocastanum	<i>Acer pseudoplatanus</i>	acer.pseudoplatanus	<i>Juglans regia</i>	juglans.regia
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Methods metadata	
General experimental design	Extracted from: Lopera Doblas (2017) Field Season Protocol -handbook.pdf There have been approximately 3700 individually marked oak trees in Silwood, divided into three categories: long term oaks which are monitored every year, odd year oaks and even year oaks which are monitored in odd and even years respectively. Trees should be marked at the beginning of the season with tape or paint in order to identify from the distance. Trees are associated by distance to a named bird box. There is a network of woodcrete nest boxes across the campus' woodlands used to study the breeding phenology of blue tits.
Data collection	Monitoring involves visiting every individual oak every few days from April onwards to look for signs of leaf development, and scoring them according to the scale below, until they reach stage 6. leaf bursting is not uniform across all branches of the tree. We find the score most of the tree branches show or, if this is difficult to assess, record the highest score we observe.  Figure: different stages of the leaves 0 = Leaf scales are close without signs of green 1 = First sign of 'leaf burst' when green is first visible between brown bud scales 2 = 'Big bud' is when the bud has elongated and is green starting to show the tip of new leaves/flowers 3 = Leaves and male flowers project beyond the tip of the bud but are still small and joined together on the base 4 = 'Leaf extension' is when individual bright green and soft leaves and anthers (male flowers) hang separately 5 = 'Anthesis' is when pollen is shed by anthers (this stage is optional) 6 = 'Full leaf expansion' is when tanning is deposit on leaves, they become dark green and hard Once a tree reaches stage six it is also given a defoliation score (how much has been eaten by caterpillars!). Defoliation scores: 6 = 100 % leaf area loss 5 = 50-99% 4 = 25-49% 3 = 12-24% 2 = 6-11% 1 = 1-5% 0 = intact leaf

	At this stage, the person recording the stage of the tree should stand under the tree to get the defoliation score. For this, divide the tree in areas and establish a general score for the whole tree. It could happen that the upper part of the tree is been eaten and the bottom has not. Girth: There is information of the circumference of some trees. Raw data file “oaks_GIS data” contains circumference information for some trees but has not a date associated. In the dataset (girth.csv) these measures have been identified as 2007-2015 girths. Some measures have been done since from a period of three years, also without a date, these are identified as 2016-2019. None have an associated VisitID. Read README_OakDataset.txt to find more.
Quality control	Phenology scores have been given by different students and researchers over the years. A complete list can be found in file: observers.csv Curation of data files and creation of metadata has been done by Catalina Estrada since January 2018. Please read README_OakDataset.txt to see specific issues and decisions.
Data table metadata	
Number of tables	14
	observers.csv
	observertypeid.csv
	markings.csv
	markingid.csv
	visits.csv
	defoliation.csv
	girth.csv
	defoliation.csv
	anthers.csv
	nestboxlocation.csv
	nestboxterritory.csv
	rotation.csv
	trees.csv
	treestate.csv
	Check OakDataSet_map.pdf to find how these tables are related

File name	Phenology2007_2024.csv		
Description	Combine information from phenology.csv and visits.csv for data on the phenology scores of oak trees		
Size	3.4MB		
Case sensitive	No		
Number or records	108766		
Number of attributes	5		
Orientation	Variables (attributes) included as columns		
Data table structure and attribute description			
Attribute name	Definition	Type	Attribute description
VisitID	Unique number given to a visit to assess phenology	Integer	Count

TreeID	Unique number given to each tree	Integer	Count
Score	Number representation for the stage of leaf flushing of the tree in a given visit/day.	String	Alphanumeric Numbers 0 to 6, sometimes along with signals < and > 0 = no sign of green 1 = green just showing 2 = budburst i.e., when the bud is elongated 3 = shaving brush leaves emerged 4 = leaves fully extended 5 = trees anthers shedding pollen 6 = leaves turned dark green and waxy (tanninised) The meaning of < and > or of scores different to these above e.g. (7 to 10 or \$) have been lost
note	Text from field data collection	String	Text
Date	Day of visit to tree	Date	dd/mm/yyyy

Data anomalies	
	README_OakDataset.txt contains information of data curation for tables including the fate of potential mistakes from original data files.

Supplemental descriptors	
Publications	
Order	
How to cite database	Schroeder, J., & Estrada Montes, C. (2025). Oak tree leaf budding phenology, Silwood Park UK [Data set]. Zenodo. https://doi.org/10.5281/zenodo.14771428
How to acknowledge dataset	Contact c.estrada@imperial.ac.uk
Additional information	Sampling protocols can be found in: Lopera Doblas (2017) Field Season Protocol, file: handbook.pdf A map linking tables by key columns: OakDataset_map.pdf Map showing nest boxes: nest boxes.jpg